

Bursting and budding pathogenesis

Depletable defences and the advantage of concentrated release

Adam Aldridge

University of Leeds

Contents

1	Bursting and budding pathogenesis	2
1.1	Introduction	2
1.1.1	Three candidate advantages	2
1.2	Why the release schedule alone cannot matter	4
1.2.1	Successive birth–death–catastrophe	4
1.3	Intracellular depletion: replicator and suppressor	6
1.3.1	The granule store	6
1.3.2	The replicator–suppressor process	7
1.3.3	First-step analysis	8
1.3.4	The survival maps	10
1.4	Extracellular depletion: fixed removal	11
1.4.1	The playing field	11
1.4.2	The replication ratio	13
1.4.3	Analysis of proliferability	15
1.5	Bursting and budding: where the advantage lives	16
1.5.1	What is settled, and what was left open	16
1.5.2	The reversal	17
1.5.3	Placement on the spectrum	18
1.5.4	<i>Y. pestis</i>	19
1.6	Discussion	20
1.6.1	Limitations	20
1.6.2	Open problems	21
1.6.3	Connections forward	21
1.A	Numerical verification	21
1.B	Computational notes	22
1.B.1	The first-step solver	22
1.B.2	Quadtree sampling	23

Chapter 1

Bursting and budding pathogenesis

1.1 Introduction

Under linear clearance the schedule by which an infected cell releases its load is invisible to the basic reproduction number: replacing constant budding by a single burst leaves R_0 exactly unchanged, a fact proved in the chapter on population dynamics (??). The schedule is not invisible to establishment, but that comparison is again a comparison of branching structures at fixed mean. This chapter asks whether a depletable defence changes either conclusion — whether bursting can be an advantage, or a disadvantage, once the host's capacity to kill is finite.

Two sites are treated. Inside the host cell, some leucocytes carry a finite store of oxidative granules, spent as they act. Outside it, killer cells in the extracellular matrix remove free agents on contact and are themselves used up. In both cases the per-agent hazard depends on how many agents arrive at once. Constant budding, as used in the basic model of viral replication [1, 2, 3], is an approximation that ignores that dependence; single-cell observations of HIV release are already intermittent rather than constant [4, 5, 6]. Eclipse phases and load-dependent production [7, 8, 9] address a different approximation, and are not pursued here.

Comparing bursting with budding is not new [10, 11, 12, 13, 14]. What is added is a quantification of *flooding* — that a burst-style release is more likely to exhaust the local killing capacity. The word is used below in two senses that must be kept apart — a comparison of two offspring laws at one delivery, and a comparison of the same yield split across more deliveries — and the two do not always agree.

1.1.1 Three candidate advantages

Is there a fundamental advantage to bursting dynamics in pathogenesis? For *F. tularensis* and *Y. pestis*, host cell lysis may be no more than the simplest available means of release: bacteria are large in comparison with the protein complexes that constitute a cell membrane, so it would be difficult to employ a strategy in which some but not all of the pathogen were released. Viruses are different. They are small enough to allow partial release of an intracellular load, and for viruses such as HIV there is an explicit advantage to not killing the cell, since its machinery continues to produce virions. That an all-at-once strategy nonetheless persists is what needs explaining, and the explanations on offer fall into three categories.

Subductive proliferation. Replication rate declines over time within a host cell, perhaps following recognition, or as the resources available there are used up.

Limited killing. The means of removing agents are finite on the relevant timescales, so the effective death rate of invading pathogen falls as the incident cohort is killed. This may happen inside the cell, where some host cells carry oxidative granules of finite number, or

outside it.

Detection avoidance. Continual budding may be detected more readily, by immune cells that induce apoptosis, than release that is sporadic and clustered, as HIV's is observed to be.

All three amount to catching some immune mechanism off guard. This chapter pursues the second, because it is the one that can be settled by calculation; the other two are named here and left. Subductive proliferation would need no new machinery, its dynamics being the standard time-inhomogeneous theory already available in ??, but that theory does not by itself distinguish bursting from budding. Detection avoidance would need a model of detection, which this chapter does not have. Section 1.6.1 returns to both.

Three results carry what follows. If a fixed count ϑ of released agents is removed on each delivery, the expected surviving release of one cell is $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$, where V_∞ is the expected lifetime yield of an infected cell and a is the larger root of the characteristic quadratic taken from the chapter on the birth–death–catastrophe process. Every unit of removal costs the burst one factor of a . In the replicator–suppressor model of section 1.3, survival is certain once the incident cohort exceeds the granule store; on the matched diagonal the infection fails with a probability that decays only polynomially in cohort size, at exponent λ/ς . And the flooding criterion of the chapter on population dynamics reverses under a removal threshold: bursting wins the matched-mean comparison exactly when $L < 1$, the complement of ??.

The derivation proceeds as follows.

- A chained birth–death–catastrophe process with complete transfer cannot prefer one release schedule, so a nonlinearity is forced into the model.
- The intracellular chain is solved by first-step analysis; the certainty threshold and the diagonal law are closed.
- The extracellular toll is reduced to an integer budget, and memorylessness of the burst law gives $\mathcal{R}(\vartheta)$.
- The same functional is compared across bursting and budding, and the criterion reverses.

The intracellular process, its roots a and b , its lifetime yield V_∞ and its burst-size law are taken from the chapter on the birth–death–catastrophe process (??????); the single-cell comparison of the two release strategies is established in the chapter on its distribution theory (??); and the population comparison, the flooding theorem and the catalogue of intermediate release mechanisms are proved in the chapter on population dynamics (????). New here are the two depletion models, the closed form $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$, and the reversal. Section 1.2 forces the nonlinearity; sections 1.3 and 1.4 supply it, one site each; section 1.5 places the result against the flooding theorem and closes on *Y. pestis*. Section 1.A records the numerical checks.

Symbol	Meaning
<i>Inherited, and used here without redefinition</i>	
λ, μ, δ	birth, death and catastrophe rates of one replicator
a, b	roots of the characteristic quadratic, $b < 1 < a$
$L = a(1 - b)$	the flooding parameter of ??
W_t	agents released by one infected cell up to time t
V_∞	expected lifetime release of one infected cell, $\lim_t \mathbb{E}(W_t)$
r	burst size: the load released at rupture
q	probability that a released agent infects a new cell
<i>Particular to this chapter</i>	
ϑ	fixed removal budget, an integer count of killed agents
ϑ^*	the budget at which $\mathcal{R} = 1$, $\log V_\infty / \log a$
$\mathcal{R}(\vartheta)$	expected surviving release per cell under budget ϑ
X_t, Q_t	intracellular replicators, and remaining granules
ς	suppression rate per replicator–granule pair
$\pi_{i,j}$	probability of extinction from i replicators, j granules
N	granule store of a fresh host cell

Table 1.1: Notation. The upper block is taken from the chapter on the birth–death–catastrophe process and the chapter on population dynamics; the lower block carries meanings specific to this chapter, and those letters were chosen to avoid collisions with the ones those meanings would otherwise have taken.

1.2 Why the release schedule alone cannot matter

In the pursuit of modelling multi-stage intracellular infection, an obvious first move is an iterated birth–death–catastrophe process with the assumption of complete transfer — chained ruptures. In trying to determine whether there is a fundamental reason for bursting dynamics in pathogenesis, this model is of limited efficacy. With agents replicating identically whenever they are within a host cell, it matters not how they are moved, just that they are replicating. With chained ruptures, the burst event is merely a recorded instance without tangible implication to agent spread. And if instead the bacteria modelled were to make their way one by one to a new host, or even to multiple hosts, no change would be made to the expected growth rate of the overall population. Even if some delay were introduced for unit transfer between hosts, this would merely have the effect of adding intervals of inactivity to the time axis; broad propagation dynamics would remain identical.

It seems, then, that if we wish to understand why pathogens follow a bursting regime of infection — or whether there is a reason at all — we must consider additional factors. Namely, we must take into account that the death and replication rates of individual agents are not entirely independent of the surrounding cohort. Various of the body’s immune factors are depletable. Leucocytes including macrophages and neutrophils contain on the order of ten so-called granules, small packets containing oxidising compounds with a toxic effect on invading pathogens. Given that these can be used up, the effective within-host death rate of infectious agent will reduce as more and more of the incident cohort are killed. There are likely extracellular immune factors which exhibit a similar pattern of depletion.

1.2.1 Successive birth–death–catastrophe

The concept of chained ruptures is simple. A series of birth–death–catastrophe processes follow on from one another, the result of the last going on to found an initial condition for the next [15, 16, 17]. Starting with a single replicative unit, the first process undergoes the standard dynamics of birth and death. Then, at the time of the catastrophe event, the quantity of units present immediately transfers into another target cell, founding the second process with its initial

condition as given. This is the main assumption: that released replicative units somehow find their way, all at once, to the same host cell. Of course this is quite reduced from the biological context. The chapter on its distribution theory solves the chain in closed form at $\mu = 0$ for every chain position, giving the rupture-size law $\mathbb{P}(r_k = n) = \binom{n+k-2}{k-1} s^k \rho^{n-1}$ of ?? together with its moments and rupture times, and records the case $\mu > 0$ as open (??).

A less restricted view would be that following rupture, all of the r_1 released bacteria undergo Brownian motion and drift gradually from the burst location. The cohort, as it spreads out, will encounter multiple new target cells and get taken up one by one. Albeit less so, this is also greatly reduced from biological reality, which in its full complexity will be beyond the scope of concise mathematical description. For example, the target cells we have focused on mostly are macrophages. Macrophages do not merely float around in Brownian drift; they are motile, and essentially walk around their environment seeking out debris and alien material to engulf and make away with. The agents themselves are little more straightforward. *Y. pestis* can “decide” whether it will be absorbed or not by such phagocytes [18, 19]. In addition to this, the bacteria can inject immune cells with a host of special proteins with various effects including neutralisation [20].

Generally, the playing field of these actors is just that, an environment of play, in which the responsive agents interact with one another; causal relationships ensue which go far beyond mere drift and chance.

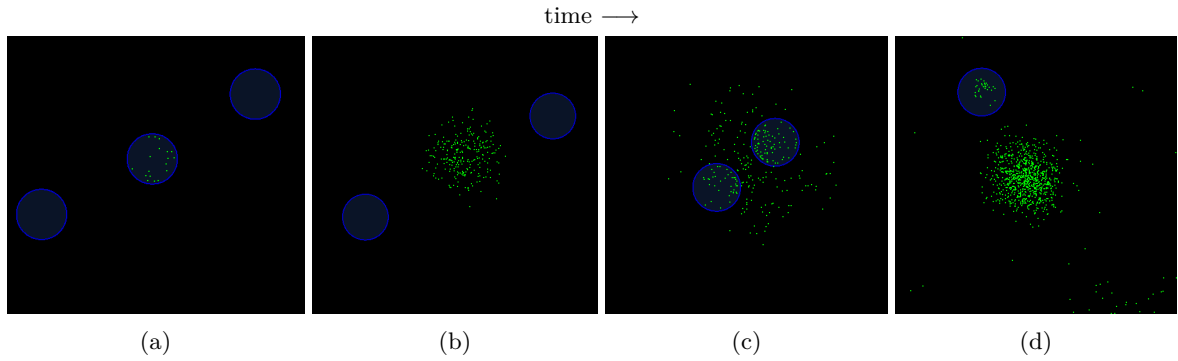


Figure 1.1: Four frames of a particle simulation, showing what the chained model leaves out. (a) Intracellular replication before bursting: the load accumulates out of reach. (b) The cell bursts and the contents span the extracellular space before finding new hosts — the interval the chained model collapses to a point. (c) Macrophages are motile; they wander the extracellular matrix phagocytosing debris and extracellular pathogens alike, so which cell a released agent reaches is not settled by diffusion alone. (d) A second round of bursting, here with a larger load. Under identical intracellular rates the growth rate is the same however the transfers between (a) and (d) are arranged, which is why the argument of this section forces a nonlinearity into the model.

Figure 1.1 shows the four stages of that picture as a simulation run. What matters for the argument is the contrast between panels (a) and (d): both show a burst, and under identical intracellular rates the two carry the same expected yield whatever route the released agents take between them.

The conclusion is negative, and it sets the chapter’s agenda. A release schedule can only matter if something in the environment responds to how many agents arrive at once — if the per-agent hazard depends on the size of the cohort it arrives in. Two such mechanisms are treated in what follows, one inside the host cell and one outside it: section 1.3 makes the intracellular killing capacity finite and depletable, and section 1.4 does the same for the extracellular one. Both are nonlinear, and both are therefore outside the scope of the branching comparison in the chapter on population dynamics, which compares offspring distributions at fixed mean and says so explicitly (??).

1.3 Intracellular depletion: replicator and suppressor

A number of replicative units found an infection within a host cell, and their dynamics accord with a continuous-time birth process; what is the probability that at time t at least one unit is still alive, so that the infection has not died out? For a linear process this depends only on the criticality — whether a given unit is expected to leave behind more or less than a replacement of itself — and $????$ settle it. $??$ asks the question this section asks, in its linear form: how much difference does the size of the founding group make? What is sought here is that same non-extinction probability through time when the dynamics are nonlinear, and with it the effect of the nonlinearity on survivability at varied initial counts $X_0 = n$.

1.3.1 The granule store

Most of the handful of leucocyte types known to science carry packets of toxic material called granules; granulocytes not least. Their contents are generally oxidative, and the immune cell employs them to attack invading pathogens [21, 22]. The leucocytes of interest in this research, macrophages and neutrophils, both carry granules, but they carry different numbers of them and use them differently. Neutrophils are of the category granulocyte and, as the name suggests, hold more granules than macrophages; they lyse themselves to release those inflammatory granules into the extracellular medium. Macrophages instead employ their granules intracellularly, merging them with the phagosomal membranes that harbour pathogen after engulfing has taken place — a compartment some pathogens escape rather than survive [23].

It is this dynamic which forms the basis of the model presented here. Picture a leucocyte which has just engulfed some number of pathogen and which holds, besides, some number of toxin-containing granules, assumed to be depleted in the process of being used to kill. Because the granules are finite in number and depletable, a larger incident cohort of replicators may have a greater chance of eventual survival. This is the intracellular form of the dose-dependence that stochastic models of *F. tularensis* and of *Salmonella* have found at the level of the whole host [24, 25, 26, 27].

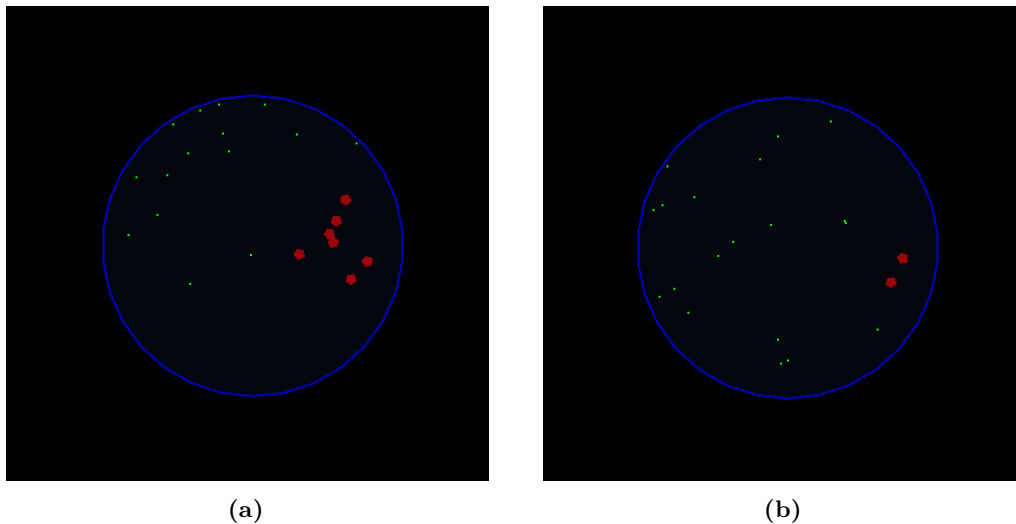


Figure 1.2: The two opposing forces, simulated. Intracellular granules are in red, pathogenic particles in green; when one of each makes contact, both are annihilated, and that mutual annihilation is the depletion the model turns on. (a) Early, with the store still nearly full. (b) Later in the same run, with most of the store spent: the pathogen count is little changed, but the suppression rate ς_{ij} has fallen with j , so what remains of the defence can no longer keep pace with the birth rate.

Figure 1.2 shows the two forces in a simulation. The point to take from it is that the granules

are consumed as they act: the red population in panel (b) is smaller than in panel (a), and it is that reduction, not any change in the pathogen, which lets the green population run away.

1.3.2 The replicator–suppressor process

We explore a simple model representing the interplay between the two opposing forces: replicative pathogenic particles fighting for survival against their host cell’s inbuilt defence mechanism. The model constitutes two unit types, and from a defined initial condition the dynamic between them results in either pathogen extinction or the overthrowing of immunity, which here appears as unbounded growth in the replicator count.

Definition 1.1 (Replicator–suppressor process). Let X_t count the infectious agents inside one host cell and Q_t the remaining number of within-cell granules. The process $\mathcal{X} = \{(X_t, Q_t) : t \geq 0\}$ is the continuous-time Markov chain on

$$\mathcal{S} = \{(i, j) : i \in \mathbb{N} \cup \{0\}, j \in \{0, \dots, N\}\} \quad (1.1)$$

with, for $(i, j) \in \mathcal{S}$,

$$\begin{aligned} \mathbb{P}(\Delta X_t = 1, \Delta Q_t = 0 \mid X_t = i, Q_t = j) &= \lambda i \Delta t + o(\Delta t), \\ \mathbb{P}(\Delta X_t = -1, \Delta Q_t = -1 \mid X_t = i, Q_t = j) &= \varsigma i j \Delta t + o(\Delta t), \\ \mathbb{P}(\Delta X_t = 0, \Delta Q_t = 0 \mid X_t = i, Q_t = j) &= 1 - (\lambda i + \varsigma i j) \Delta t + o(\Delta t). \end{aligned} \quad (1.2)$$

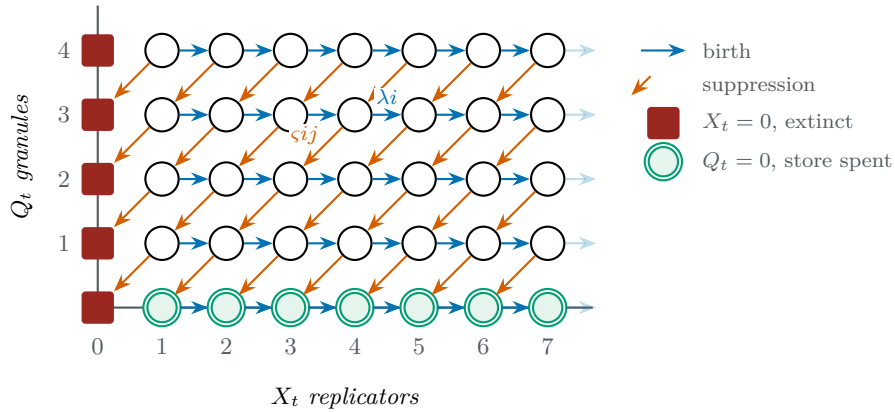


Figure 1.3: The state space $\mathcal{S}$ of definition 1.1. Births move right and spend no granule; suppressions move down and left, spending one of each. The column $X_t = 0$ is extinction; the row $Q_t = 0$ is a spent store, from which only births remain, so survival is certain. That is why the diagonal $i = j$ matters.

Units of X_t divide and increase according to the standard mechanism of a birth process. Suppressive interactions occur at a rate proportional to the product of the two counts and result in a single elimination of each type; it is that product which makes the process nonlinear, and it is the only nonlinearity in the model. Because each suppression costs one of each, a cohort that outnumbers the store cannot be cleared.

In reality growth will not be truly without limit — intracellular capacity will eventually impose a threshold on further expansion without cell bursting — but the model is not run beyond the point at which the granules are exhausted, so the idealisation costs nothing that is used below. Figure 1.3 is the chain.

Our interest is in the relationship between incident pathogen count and infection survivability, that is in

$$\mathbb{P}(\text{infection survives}) = 1 - \lim_{t \rightarrow \infty} \mathbb{P}(X_t = 0), \quad (1.3)$$

which is the probability of late-time non-extinction. Survival is assured on any path that exhausts the antagonistic granules: once $Q_t = 0$ the process is a pure birth process, and extinction is impossible thereafter.

Remark 1.2 (No replenishment). The granule store is fixed at Q_0 and never supplemented. One may devise a model in which granules are produced at some rate by the cell, and for a long infection that would be the right model; the version analysed here takes the initial immune capacity as given, which is the regime in which a single engulfment episode is decided. Every result below is a statement about that regime, and the certainty threshold of [proposition 1.3](#) in particular depends on it.

The asymmetry of the two arrow families is the whole model: a birth spends nothing, whereas a suppression spends a granule, so the store can only fall and each unit of it can be used once.

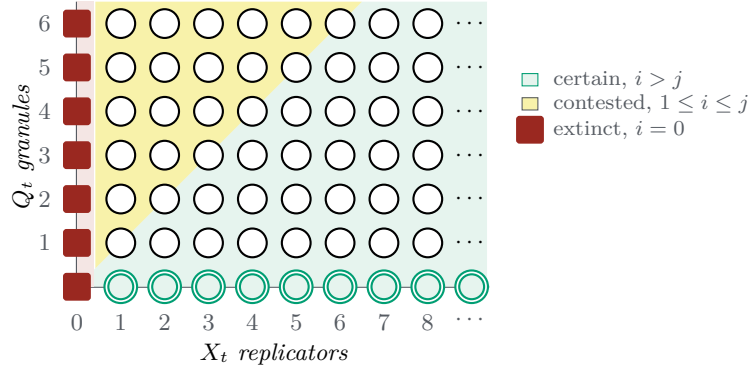


Figure 1.4: The three fates, read off [figure 1.3](#). In $i > j$ survival is certain: extinction would need i suppressions and only j are available ([proposition 1.3](#)). In $1 \leq i \leq j$ both outcomes have positive probability, and $\pi_{i,j}$ is what the rest of this section computes. The boundary is the diagonal, and it is sharp.

[Figure 1.4](#) is the section's headline. A cohort that outnumbers the store cannot be cleared, however unfavourable the rate constants; a cohort that does not outnumber it may still win, and with what probability is the question.

1.3.3 First-step analysis

The process is defined by its one-step transition probabilities,

$$p_{(i,j) \rightarrow (k,l)}(\Delta t) = \mathbb{P}((X_{t+\Delta t}, Q_{t+\Delta t}) = (k, l) \mid (X_t, Q_t) = (i, j)),$$

and enumerating the steps available from an arbitrary state (i, j) gives

$$p_{(i,j) \rightarrow (k,l)}(\Delta t) = \begin{cases} \lambda i \Delta t + o(\Delta t), & (k, l) = (i + 1, j), \\ \varsigma i j \Delta t + o(\Delta t), & (k, l) = (i - 1, j - 1), \\ 1 - (\lambda i + \varsigma i j) \Delta t + o(\Delta t), & (k, l) = (i, j), \end{cases} \quad (1.4)$$

as $\Delta t \rightarrow 0$. Write $\pi_{i,j}$ for the probability of fixation in the set $\{X_t = 0\}$ from the state (i, j) . Conditioning on the first jump, and using the first-step principle of ?? in the continuous-time form ??,

$$\pi_{i,j} = \hat{\alpha}_j \pi_{i+1,j} + \hat{\beta}_j \pi_{i-1,j-1}, \quad \hat{\alpha}_j = \frac{\lambda}{\lambda + \varsigma j}, \quad \hat{\beta}_j = \frac{\varsigma j}{\lambda + \varsigma j}, \quad (1.5)$$

with $\pi_{0,j} = 1$ for every j and $\pi_{i,0} = 0$ for every $i \geq 1$. The jump probabilities depend on j alone: the competing rates are λi and $\varsigma i j$, and the common factor i cancels. The replicator count

therefore affects the outcome only through how many suppressions are needed, never through how likely the next event is to be one.

Collecting (1.5) over all transient states gives the linear system

$$\mathbf{p}^{(0)} = \mathbf{A} \mathbf{p}^{(0)} + \mathbf{c}, \quad (1.6)$$

in which $\mathbf{p}^{(0)}$ collects the unknowns $\pi_{i,j}$ for $1 \leq i \leq j \leq N$, the matrix $\mathbf{A}$ carries the coefficients $\hat{\alpha}_j$ and $\hat{\beta}_j$, and the vector $\mathbf{c}$ carries the single inhomogeneous term $\hat{\beta}_j$ contributed by $\pi_{0,j-1} = 1$ at $i = 1$. The system is block lower-triangular when the states are ordered by the level $j - i$, because the two states appearing on the right of (1.5) sit one at level $j - i - 1$ and one at level $j - i$; those levels are drawn in figure 1.5. Nothing beyond that structure is needed: it makes the system solvable level by level, and the solver of section 1.B exploits nothing else.

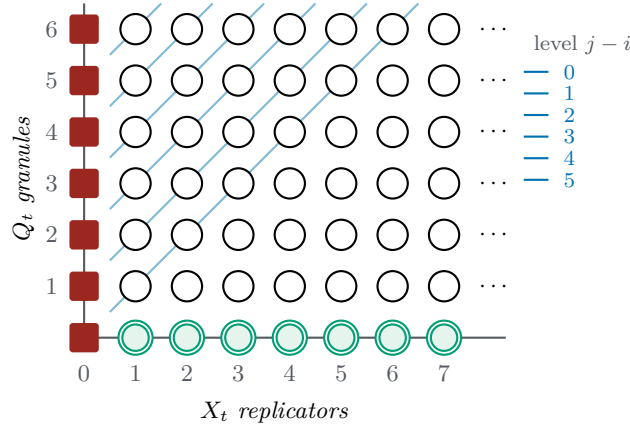


Figure 1.5: The level structure of the first-step system (1.5). States are grouped by the level $j - i$. A suppression stays on its diagonal; a birth moves one level down. No transition raises the level, so $\mathbf{A}$ in (1.6) is block lower-triangular in this ordering and the levels may be solved in turn, starting from level 0.

Level 0 is the diagonal $i = j$, and it closes on itself. That is enough for two results.

Proposition 1.3 (Certainty threshold). $\pi_{i,j} = 0$ whenever $i > j$: the infection survives with probability one as soon as the incident cohort exceeds the granule store. For $1 \leq i \leq j$ we have $\pi_{i,j} \in (0, 1)$.

Proof. Each suppression removes one pathogen and one granule; a birth removes no granule. Reaching $X = 0$ from (i, j) therefore requires at least i suppressions, so $j \geq i$ as claimed.

Now take $1 \leq i \leq j$. A run of i suppressions reaches $(0, j - i)$ and has positive probability, since $\hat{\beta}_j > 0$ at every $j \geq 1$; thus $\pi_{i,j} > 0$. A run of $j - i + 1$ births reaches $(j + 1, j)$, where $\pi = 0$ by the first claim, and has positive probability since $\hat{\alpha}_j > 0$; thus $\pi_{i,j} < 1$. $\square$

The boundary is the diagonal drawn in figure 1.4, and nothing about the rate constants moves it. Consequently the only remaining question on the contested side is how $\pi_{i,j}$ varies with dose. Monotonicity in i and in j was checked numerically over $\varsigma \in \{0.1, 0.2, 0.3\}$ at $N = 20$, and both hold everywhere (section 1.A).

Proposition 1.4 (The diagonal law). Write $\nu = \lambda/\varsigma$. On the matched diagonal,

$$\pi_{j,j} = \prod_{m=1}^j \frac{\varsigma m}{\lambda + \varsigma m} = \frac{j! \Gamma(\nu + 1)}{\Gamma(j + \nu + 1)} = \binom{j + \nu}{j}^{-1} \sim \Gamma(\nu + 1) j^{-\nu} \quad (j \rightarrow \infty), \quad (1.7)$$

so a cohort matched to its own granule store fails with a probability falling only polynomially in cohort size, at exponent λ/ς . When $\lambda = \varsigma$ this is exactly $1/(j + 1)$.

ς	$\nu = \lambda/\varsigma$	$\pi_{5,5}$	$\pi_{10,10}$	$\pi_{25,25}$
0.1	10	3.33×10^{-4}	5.41×10^{-6}	5.45×10^{-9}
0.2	5	3.97×10^{-3}	3.33×10^{-4}	7.02×10^{-6}
0.3	10/3	1.34×10^{-2}	2.26×10^{-3}	1.54×10^{-4}
1.0	1	1.67×10^{-1}	9.09×10^{-2}	3.85×10^{-2}

Table 1.2: The diagonal law (1.7) at $\lambda = 1$. Reading across a row shows the decay in cohort size at fixed rates; reading down a column shows how much slower that decay becomes as suppression strengthens relative to birth. The bottom row is the borderline case $\lambda = \varsigma$, where the failure probability is exactly $1/(j+1)$ — a cohort of twenty-five matched to twenty-five granules still fails one time in twenty-six.

Proof. From (j, j) a birth reaches $(j+1, j)$, where $\pi = 0$ by proposition 1.3. Hence $\pi_{j,j} = \hat{\beta}_j \pi_{j-1,j-1}$ with $\hat{\beta}_j = \varsigma j/(\lambda + \varsigma j)$, and $\pi_{0,0} = 1$. Iterating gives the product, and the ratio of Gamma functions follows from $\prod_{m=1}^j \varsigma m = \varsigma^j j!$ and $\prod_{m=1}^j (\lambda + \varsigma m) = \varsigma^j \Gamma(j + \nu + 1)/\Gamma(\nu + 1)$. The asymptotic is the standard ratio limit $\Gamma(j+1)/\Gamma(j+\nu+1) \sim j^{-\nu}$. $\square$

Table 1.2 shows why (1.7) is worth stating separately. Doubling the cohort on the diagonal does not halve the failure probability: it divides it by 2^ν , and $\nu = \lambda/\varsigma$ is a property of the cell rather than of the dose. Safety in numbers is therefore real but slow, and qualitatively weaker than the certainty that arrives the moment the cohort crosses the store.

1.3.4 The survival maps

Solving (1.6) over the whole grid gives the survival probability $1 - \pi_{i,j}$ for every initial condition.

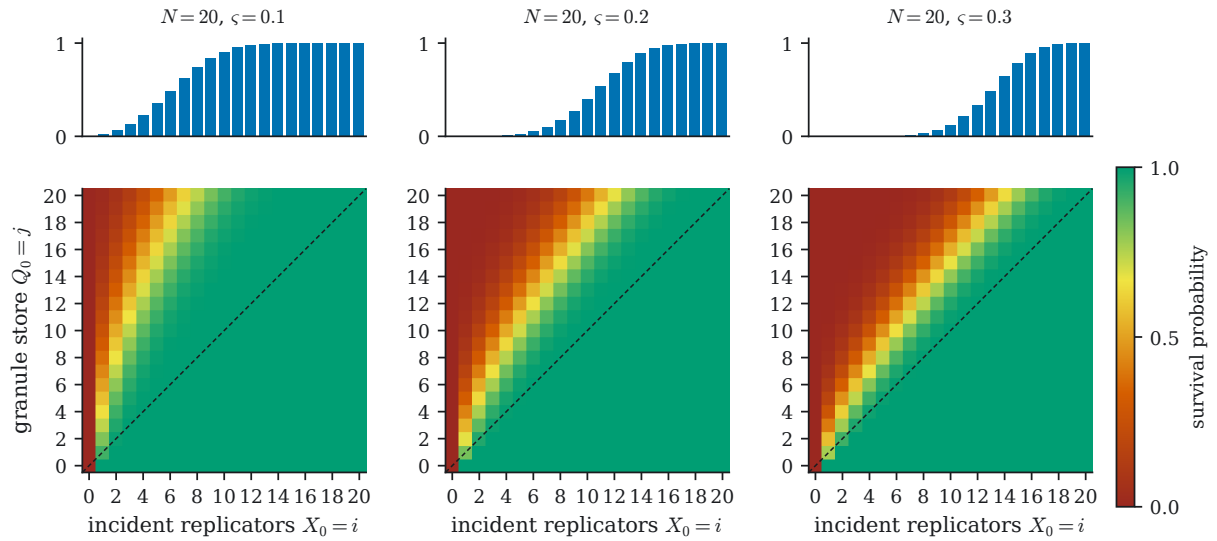


Figure 1.6: Survival probability $1 - \pi_{i,j}$ at $\lambda = 1$ and $N = 20$, for $\varsigma = 0.1, 0.2$ and 0.3 (left to right). Each lower panel is the full grid of initial conditions, with the dashed line marking the diagonal $i = j$; each upper panel is the marginal at a full store, $1 - \pi_{i,N}$, which is the case a freshly engulfing macrophage presents. Everything below the diagonal is certain survival (proposition 1.3); the transition above it steepens as ς falls, so that a weakly armed cell is not merely less effective but less discriminating about dose.

Figures 1.6 and 1.7 show the maps at $N = 20$ and $N = 10$. Two features carry the section. The first is the flat region of certain survival below the diagonal, which is proposition 1.3 drawn: it is not that survival becomes likely there, but that it becomes certain. The second is the marginal bar chart above each map, the case $j = N$ of a full store, in which the rise from

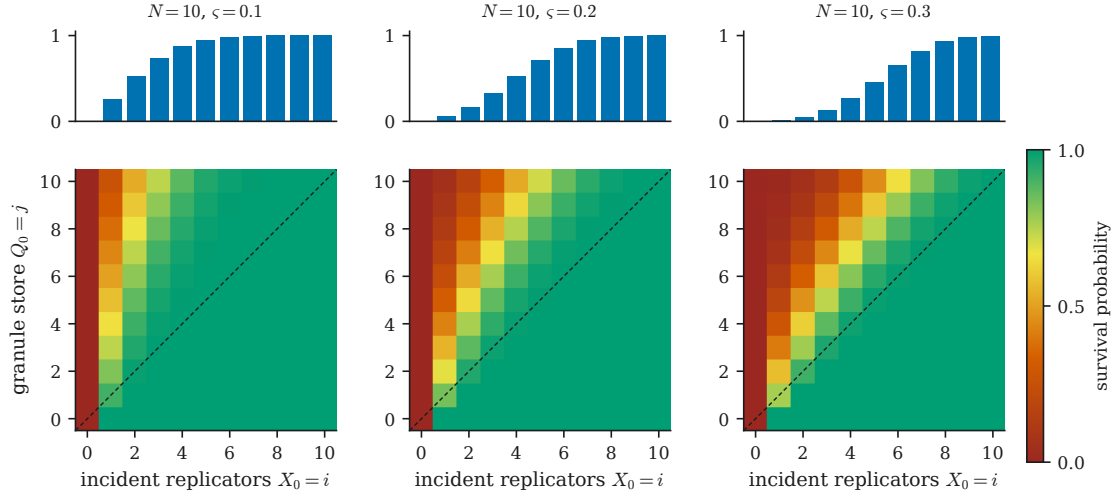


Figure 1.7: The same quantity at $N = 10$, again with $\lambda = 1$ and $\varsigma = 0.1, 0.2, 0.3$. Halving the store does not halve the dose a pathogen needs: the certainty boundary moves with N , so the requirement is a matter of the ratio of cohort to store rather than of the cohort alone.

near-zero to near-one is spread over several units of dose rather than concentrated at one — the polynomial decay of (1.7), seen from the side.

For a pathogen, the reading is direct. If a host cell can be entered by several agents at once rather than one at a time, and the store is finite, then arriving together is worth more than arriving in succession by exactly the margin these maps display. That is the intracellular version of the flooding advantage, and section 1.4 builds its extracellular counterpart.

1.4 Extracellular depletion: fixed removal

The second nonlinearity is extracellular, and it enters as an extension to the chained-rupture model of section 1.2.1. The extension is one assumption, arising from the same argument of immune depletion: that following a burst release event, some defined fixed count of the agents is removed. From that assumption alone, the question of whether bursting strategies might be of fundamental advantage can be put to a calculation.

1.4.1 The playing field

Consider again the concept of spatial drift, which for all its imperfection improves on the simpler dynamic of chained immediate transfer. Picture the playing field: a three-dimensional extracellular matrix populated by three types of agent — replicative units, that is pathogen; host cells, here macrophages; and another type of immune-active cell, which we shall call simply *killer cells*. This third type is able to eliminate units of pathogen on contact, being used up in the process: one-to-one encounters in which a killer cell removes a replicative agent and itself from play.

To begin with, the extracellular matrix is populated evenly with target cells and killer cells, each group occupying the space at a uniform spatial density. Killer cells are smaller and more numerous than target cells, a greater number per unit volume. Dynamics within target cells are as understood in the birth–death–catastrophe process: the units of pathogen replicate and die within the cell until eventually rupture is caused and the load is released.

In this model, movement is taken to be Brownian. An extracellular colony will drift from its burst location, spreading out and coming into contact with surrounding cells. Some will encounter target cells, others killer cells. The intention here is not to construct a full mathematical

description of this game, which would be cumbersome and intractable; it can be simulated computationally, and that simulation is what supports the assumption the model is used to justify. Suppose that no new killer cells are introduced into the environment, so that pathogen movement and absorption occur on a shorter timescale than killer generation. Then, with greater burst sizes, each individual bacterium will have a smaller probability of being removed. Those at the leading edge, as it were, clearing the way for their comrades in the rear. Figure 1.8 puts a large and a small release of that simulation side by side.

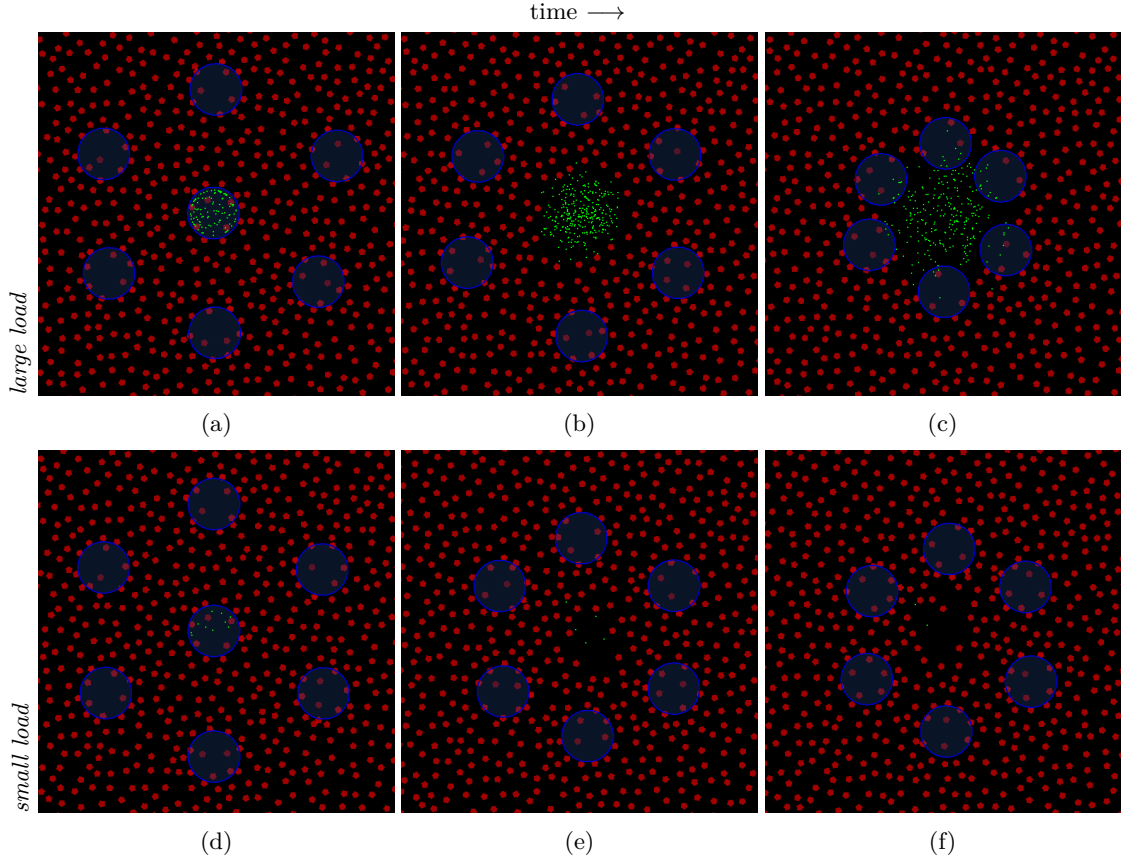


Figure 1.8: Two runs of the same simulation differing in released number alone; every rate is the same. Depletable extracellular immune factors are consumed by the agents they remove, so a large load creates a safety in numbers — a “flooding effect” — and the killers that act early are not available later. The comparison to draw is between (c) and (f). In (c) the large load has cleared a patch of the killer field and survivors remain inside it, poised on the surrounding host cells; in (f) the small load is exhausted before the field is, and nothing persists to the next round of absorption and replication.

If free agents do tend to find a new host before long — an assumption taken to its limit in the case of chained immediate transfer — then this observation offers the basis of an approximate mathematical rule. Burst size and the number of pathogen killed in the process of transfer stand in some relation to one another, and across larger and larger bursts the casualty count should rise ever more slowly. Qualitatively this might be a logarithmic progression,

$$\vartheta(r) \sim \log(r + 1), \quad (1.8)$$

where r is a given burst size and ϑ the number of replicative agents removed by killer cells; the $+1$ simply makes the curve pass through the origin.

Figure 1.9a draws that curve, and its flatness is what the rest of the section idealises. It is reasonable to impose a *fixed removal* following release, in which the number removed by killer cells is a specified constant regardless of burst size, except that if the burst size is smaller than

that constant, all of the units are killed. Taking ϑ now to be that constant,

$$\text{removal} = \begin{cases} r, & r \leq \vartheta, \\ \vartheta, & r > \vartheta, \end{cases} \quad \text{that is, } \min(r, \vartheta). \quad (1.9)$$

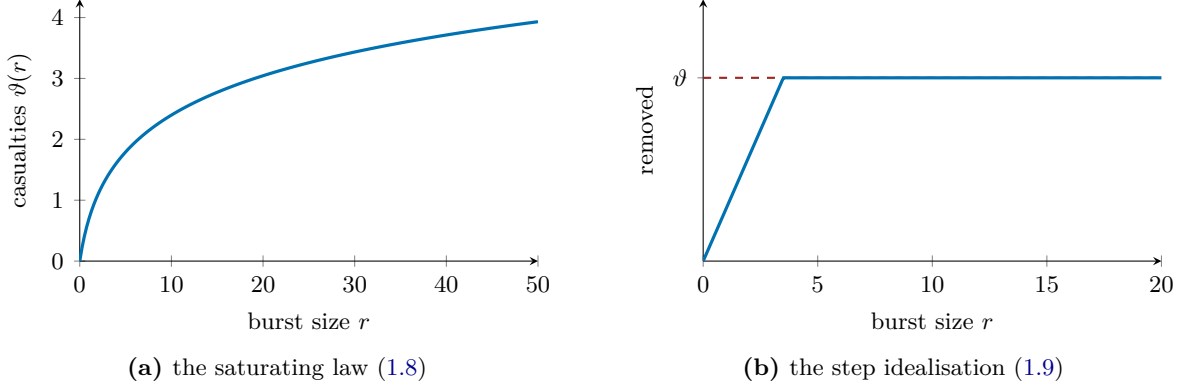


Figure 1.9: The casualty count against burst size: the law the biology suggests, and the caricature the calculation uses. In (a) the vertical axis is the number of agents killed, not a rate — what falls as the burst grows is the killed *fraction*, since the curve flattens while the burst size does not. Doubling a load of ten adds well under one casualty, which is the whole of the flooding intuition in one line. In (b) that flatness is pushed to its limit: below the budget every agent dies, and above it the plateau at ϑ means a burst pays the same toll however large it is, so all the marginal yield survives. The reason for adopting the caricature is that it makes the replication ratio computable in closed form.

Figure 1.9b shows the idealisation against which the rest of the section works. Its efficacy rests on the expectation that the same qualitative dynamics are shared with logarithmic removal, or with whatever the actual relation turns out to be, provided it is of this sort. Two further assumptions are needed before the calculation can proceed, and both are restrictions rather than conveniences.

Remark 1.5 (Killer prevalence resets). At the time of each rupture event the killer cell prevalence is assumed to have returned to its steady state and to be once again of uniform spatial distribution. This is what makes ϑ the same at every delivery. It requires the interval between ruptures to be long relative to the recovery of the effector population, whose turnover in an established infection is itself a modelled quantity [28]; where it is not, the budget paid at successive deliveries would be correlated, and the independence used below would fail.

Remark 1.6 (One target cell, or many). The fixed-removal dynamic may be framed in the context of chained transfers, where all agents released have the same target cell to go to; or in a case where each successful pathogenic unit goes on to found an infection in a new target cell. Under the first framing the replication ratio would be the expected number of surviving released units *per incident agent*. The second framing is taken here, so that the ratio counts newly infected target cells, which is what makes it comparable with the reproduction numbers of the chapter on population dynamics.

1.4.2 The replication ratio

Take, then, the second option: each released agent goes on to found a new infection in a new target cell. The quantity of interest is the effective *replication ratio* between target cell infections — the expected number of newly infected target cells created by a host founded from one initial bacterium — and because each surviving released agent infects a new host, that ratio is simply the expected number of survivors.

Define r as the outcome of a birth–death–catastrophe process with a single incident unit. We must choose whether to include outcomes that are zero through internal death, and it makes practical sense here not to ignore the cases $r = 0$; with that choice

$$\mathbb{E}(r) = \lim_{t \rightarrow \infty} \mathbb{E}(W_t) = V_\infty, \quad (1.10)$$

the lifetime yield of ???. The letter r is that of the chapter on its distribution theory as well: its r_k is the load at the k th rupture of a chain (??), and the r here is that chain’s first term.

The number surviving fixed removal is $\max(0, r - \vartheta)$, so the expected replication ratio is

$$\mathcal{R}(\vartheta) = \mathbb{E}(\max(0, r - \vartheta)). \quad (1.11)$$

This is not the R_0 of the chapter on population dynamics, and [section 1.5.1](#) states the relation between them; the change of letter is deliberate.

Conditioning on whether the burst size exceeds the budget gives

$$\mathbb{E}(\max(0, r - \vartheta)) = [\mathbb{E}(r \mid r \geq \vartheta) - \vartheta] \mathbb{P}(r \geq \vartheta), \quad (1.12)$$

the term for $r < \vartheta$ vanishing because the surviving count is then zero. Extracting $\mathbb{E}(r \mid r \geq \vartheta)$ from the decomposition

$$\mathbb{E}(r) = V_\infty = \mathbb{E}(r \mid r \geq \vartheta) \mathbb{P}(r \geq \vartheta) + \mathbb{E}(r \mid r < \vartheta) \mathbb{P}(r < \vartheta)$$

and cancelling $\mathbb{P}(r < \vartheta)$ against the denominator inside $\mathbb{E}(r \mid r < \vartheta)$ turns (1.12) into a difference of two partial sums. This is analytically computable because the burst-size law is known: unconditionally, including the mass b of realisations that never burst,

$$\mathbb{P}(r = i) = \frac{\delta}{\lambda} a^{-i} \quad (i \geq 1), \quad \mathbb{P}(r = 0) = b, \quad (1.13)$$

which is ??. Hence

$$\mathbb{E}(\max(0, r - \vartheta)) = V_\infty - \mathcal{M} - \vartheta(1 - F(\vartheta)), \quad (1.14)$$

where

$$F(\vartheta) = \mathbb{P}(r < \vartheta) = b + \sum_{i=1}^{\vartheta-1} \frac{\delta}{\lambda} a^{-i}, \quad \mathcal{M} = \sum_{i=1}^{\vartheta-1} i \frac{\delta}{\lambda} a^{-i}. \quad (1.15)$$

Because the formula $\mathbb{P}(r = i) = (\delta/\lambda)a^{-i}$ does not apply at $i = 0$, the sum for F begins at 1 and carries the separate term b in front of it: that is the probability that the process ends in internal death, an output of zero.

1.4.2.1 The law is normalised, and the tail is free

Before analysing (1.14) it is worth confirming that (1.13) is a probability distribution, since the two pieces come from different places. The probability that $r = 0$ is b , the smaller root of the characteristic quadratic,

$$b = \eta - \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad \eta = \frac{\lambda + \mu + \delta}{2\lambda}, \quad (1.16)$$

as fixed by ??, where $b < 1 < a$, $ab = \mu/\lambda$ and $a + b = (\lambda + \mu + \delta)/\lambda$. It remains to show that $b + \sum_{i \geq 1} \mathbb{P}(r = i) = 1$, that is that

$$\sum_{i=1}^{\infty} \frac{\delta}{\lambda} \left(\frac{1}{a}\right)^i = 1 - b.$$

Using $\sum_{k \geq 1} y^k = y/(1-y)$ for $|y| < 1$, the left-hand side is $\frac{\delta}{\lambda} \cdot \frac{1/a}{1-1/a}$, so the claim is equivalent to

$$\frac{\delta}{\lambda} = (1-b)(a-1) = (a+b) - ab - 1,$$

and substituting the two Vieta identities gives

$$(a+b) - ab - 1 = \frac{\lambda + \mu + \delta}{\lambda} - \frac{\mu}{\lambda} - 1 = \frac{\delta}{\lambda},$$

as required. The same identity delivers the upper tail with no further work:

$$1 - F(\vartheta) = \mathbb{P}(r \geq \vartheta) = (1-b) a^{-(\vartheta-1)}, \quad (1.17)$$

which is exact for every integer $\vartheta \geq 1$. The tail of the burst law is geometric, and (1.17) is the fact from which the next subsection's closed form follows.

1.4.3 Analysis of proliferability

The question is whether $\mathcal{R}(\vartheta)$ is greater or less than one, and if greater, by how much; and how that changes with δ . Do larger burst sizes make an infection more viable? The geometric tail of (1.17) already answers: a budget spent against a geometric law removes a fixed proportion of it rather than a fixed amount.

Proposition 1.7 (Fixed-removal replication ratio). *For an integer removal budget $\vartheta \geq 0$,*

$$\mathcal{R}(\vartheta) = \mathbb{E}(\max(0, r - \vartheta)) = V_{\infty} a^{-\vartheta}. \quad (1.18)$$

Each unit of removal costs the burst a factor $1/a$, independently of ϑ , of μ , and of everything about the burst-size law except a .

Proof. The burst law (1.13) is geometric on $\{1, 2, \dots\}$ with ratio $1/a$, hence memoryless above any integer threshold:

$$\sum_{i > \vartheta} (i - \vartheta) \frac{\delta}{\lambda} a^{-i} = \frac{\delta}{\lambda} a^{-\vartheta} \sum_{k \geq 1} k a^{-k} = a^{-\vartheta} \mathbb{E}(r) = V_{\infty} a^{-\vartheta}. \quad \square$$

Two consequences follow in one line each, and one restriction.

Corollary 1.8 (Threshold). $\mathcal{R}(\vartheta) > 1$ if and only if $\vartheta < \vartheta^* := \log V_{\infty} / \log a$. At $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ this gives $\vartheta^* = 44.13$; at $(1, 0.5, 1/3)$ it gives $\vartheta^* = 1.71$.

Corollary 1.9 (Why removal creates a threshold at all). $V_{\infty} > 1$ for every admissible (λ, μ, δ) , so $\mathcal{R}(0) > 1$ always. The subcritical regime exists only because of the removal.

Remark 1.10 (Partial-sum expansion). The same quantity expanded without memorylessness is

$$\mathbb{E}(\max(0, r - \vartheta)) = V_{\infty} - \vartheta(1-b) + \frac{\delta}{\lambda} \left(\frac{\frac{1}{a} \left(\vartheta - 1 - \vartheta \frac{1}{a} + \left(\frac{1}{a} \right)^{\vartheta} \right)}{\left(1 - \frac{1}{a} \right)^2} \right), \quad (1.19)$$

which follows from closing the sums in (1.15) by the identities $\sum_{i=1}^{n-1} s^i = s(1-s^{n-1})/(1-s)$ and $\sum_{i=1}^{n-1} i s^i = s(1-n s^{n-1} + (n-1)s^n)/(1-s)^2$. It is exact at integer ϑ and is recorded because section 1.A checks it independently of (1.18).

Remark 1.11 (Scope: ϑ is an integer). The derivation sums to $\vartheta - 1$ and so requires $\vartheta \in \mathbb{Z}_{\geq 0}$. That is the honest reading in any case, since ϑ counts killed particles, and it is adopted as part of the model definition. For a real budget the corresponding quantity is $\mathbb{E}(\max(0, r - \vartheta)) = a^{-\lceil \vartheta \rceil} (V_\infty + (\lceil \vartheta \rceil - \vartheta)L)$ with $L = a(1 - b)$, which reduces to (1.18) exactly when ϑ is an integer; section 1.A records its accuracy.

The behaviour of (1.18) in δ is drawn in figure 1.10. The reading that matters for what follows is the geometry of the crossing: because $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$ is a product of a term that grows as δ falls and a penalty that is exponential in the budget, an increase in the toll can always be answered by a large enough burst, and the exchange rate between the two is exactly one factor of a per unit of ϑ .

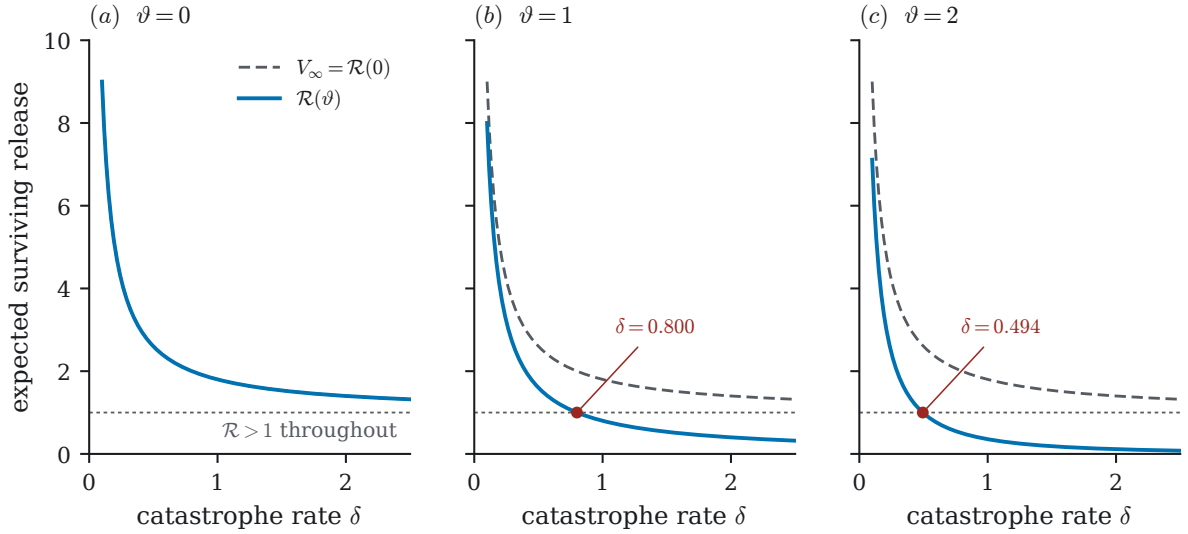


Figure 1.10: The replication ratio (1.18) against δ at $\lambda = 0.8$, $\mu = 0$, for budgets $\vartheta = 0, 1$ and 2 , with V_∞ dashed for reference throughout. In panel (a) the two curves coincide, since at zero removal the surviving release is the whole release; that panel has no crossing, which is corollary 1.9 drawn. Elsewhere the marked point is the crossing $\mathcal{R} = 1$ of corollary 1.8, at $\delta = 0.800$ and $\delta = 0.494$: to the left of it the infection proliferates, to the right the removal extinguishes it. Each further unit of budget moves the crossing left, so a pathogen facing a heavier toll needs a lower catastrophe rate — a larger burst — to stay supercritical.

1.5 Bursting and budding: where the advantage lives

1.5.1 What is settled, and what was left open

The chapter on population dynamics settles the comparison under linear clearance, and it does so twice. First, at fixed total per-cell output V_∞ , replacing constant budding by bursting leaves the basic reproduction number exactly unchanged (??): under linear clearance the release schedule is invisible to R_0 . Second, the schedule is nevertheless not invisible to establishment. ?? shows that bursting strictly lowers the extinction probability of a small inoculum, relative to matched-mean budding, if and only if the flooding parameter

$$L := a(1 - b) = a - \frac{\mu}{\lambda} \quad (1.20)$$

exceeds one, equivalently $1/\delta < 1/\lambda + 1/\mu$ (??); at $L = 1$ the two offspring laws coincide exactly and every functional of them ties (??).

The letter R_0 in the chapter on population dynamics denotes the expected number of secondary infections produced by one infected cell in a wholly susceptible population under linear clearance,

in the sense of [29]; a theorem there proves that quantity invariant. The quantity of (1.18) is not that: it is the expected number of *surviving* released agents per infected cell when the clearance is a threshold rather than a rate. Under linear clearance at infection probability q the two agree up to the thinning factor, since $\mathcal{R}(0) = V_\infty$ and the mean offspring number in the chapter on population dynamics is qV_∞ . Under threshold removal they part company, and hence the different letter. Reading $\mathcal{R}(\vartheta)$ as R_0 would put this chapter in contradiction with ??, and there is none.

?? records that both results in the chapter on population dynamics compare branching structures at fixed offspring mean, and therefore say nothing about nonlinear effects. Fixed removal is exactly such an effect — a threshold is not a rate — so the comparison can be run again through it. It does not come out the same way.

1.5.2 The reversal

Two arguments will appear, and they are not the same argument. The distributional one holds the number of deliveries fixed at one and compares two laws for that delivery, matched at mean V_∞ ; its verdict depends on L . The batch-count one holds the law fixed and asks how the same total fares when split across more deliveries, each paying a toll ϑ ; its verdict is unconditional concentration. The reversal below is the first. The spectrum figure is the second.

Proposition 1.12 (The flooding criterion reverses under threshold removal). *Let the budding comparator of ?? release a total that is geometric on $\{0, 1, 2, \dots\}$ with the same mean V_∞ , and write $\sigma = V_\infty/(1 + V_\infty)$. Then under an integer removal budget ϑ ,*

$$\mathcal{R}_{\text{burst}}(\vartheta) = V_\infty a^{-\vartheta}, \quad \mathcal{R}_{\text{bud}}(\vartheta) = V_\infty \sigma^\vartheta, \quad (1.21)$$

so bursting exceeds matched-mean budding if and only if $1/a > \sigma$, that is if and only if $V_\infty(a-1) < 1$, that is if and only if $L < 1$ — the exact complement of ??. At $L = 1$ the two laws coincide by ?? and every functional ties.

Proof. Both expressions in (1.21) follow from the memorylessness argument of proposition 1.7, applied to a geometric law on $\{1, 2, \dots\}$ with ratio $1/a$ in the first case and to one on $\{0, 1, 2, \dots\}$ with ratio σ in the second. For the criterion, take proposition 1.7 at $\vartheta = 1$ against the tail (1.17): $V_\infty/a = \mathbb{E}(r) - \mathbb{P}(r \geq 1) = V_\infty - (1 - b)$, whence $V_\infty = a(1 - b)/(a - 1) = L/(a - 1)$. Therefore

$$\frac{1}{a} > \frac{V_\infty}{1 + V_\infty} \iff 1 + V_\infty > aV_\infty \iff V_\infty(a - 1) < 1 \iff L < 1. \quad \square$$

The result is a reversal rather than a contradiction, and the reason is a single fact about variance. At matched mean, extinction probability rewards *low* offspring variance, whereas $\mathbb{E}(\max(0, r - \vartheta))$ is a convex functional of the offspring count and so rewards *high* variance. ?? gives the variance ordering in terms of the same parameter, $\text{Var}(\cdot)_{\text{bud}} - \text{Var}(\cdot)_{\text{burst}} = 2q^2V_\infty(L - 1)/(a - 1)$. The sign of $L - 1$ therefore drives both criteria, and drives them opposite ways: where bursting is the safer bet for establishment it is the worse bet against a threshold, and conversely. Both criteria turn on the offspring count being integer-valued. Only an integer count lets a threshold bite, and the same feature governs establishment [30].

Table 1.3 and figure 1.11 exhibit the reversal. The boundary is unmoved by ϑ . Carry that forward: the removal budget sets how much the winner wins by, and the rate constants alone set who wins.

If the toll is paid *per delivery*, a cell that delivers a total yield V_∞ in n instalments retains $\max(0, V_\infty - n\vartheta)$, against a bursting cell's $\max(0, V_\infty - \vartheta)$. Since $n \geq 1$, concentration wins for every n , every ϑ and every L : there is no threshold and no reversal in this version. That is the batch-count argument of section 1.4.1, not proposition 1.12.

(λ, μ, δ)	L	ϑ	$\mathcal{R}_{\text{burst}}$	$\mathcal{R}_{\text{bud}}$	winner under	
					linear clearance	threshold removal
(1, 0, 0.1)	1.100	5	6.830	7.120	bursting	budding
(1, 0.5, 1/3)	1.000	5	0.263	0.263	tie	tie
(1, 0.9, 0.1)	0.416	5	0.333	0.078	budding	bursting
(1, 0.2, 0.05)	0.862	10	7.692	7.011	budding	bursting
(1, 0.9, 0.02)	0.207	5	1.164	0.241	budding	bursting
(1, 0.95, 0.5)	1.017	3	0.138	0.142	bursting	budding

Table 1.3: The reversal in six regimes. The last two columns name which strategy wins under linear clearance and which under threshold removal, and they disagree in every row except the tie. The middle row is the boundary $L = 1$, where the bursting offspring law is exactly geometric (??) and the two ratios agree to the digits shown. The fourth row is strongly supercritical on both sides, so the reversal is not an artefact of one of them being near extinction.

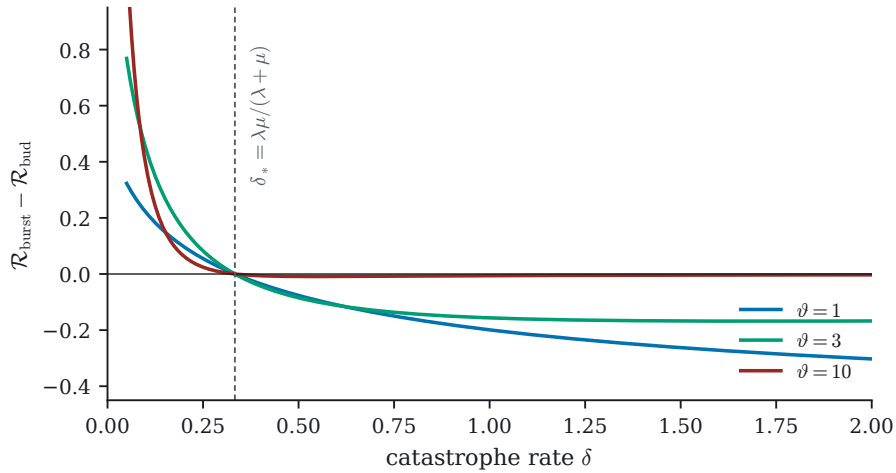


Figure 1.11: The sign change, drawn. The gap $\mathcal{R}_{\text{burst}} - \mathcal{R}_{\text{bud}}$ against δ at $\lambda = 1$, $\mu = 0.5$, for three removal budgets. All three curves cross zero at the same place, the flooding boundary $\delta_* = \lambda\mu/(\lambda + \mu) = 1/3$ of ??, and they cross it in the direction opposite to ??: bursting wins to the *left*, where $L < 1$. Raising the budget deepens the advantage on the left without moving the boundary, because the boundary is a property of the two laws and not of the toll.

Remark 1.13 (Scope of the comparison). The two arguments answer different questions and neither subsumes the other. The distributional one holds the number of deliveries fixed at one and asks which *law* for that delivery does better, and its answer depends on L . The batch-count one holds the law fixed and asks how the same total fares when split across more deliveries, and its answer does not depend on L at all. Conflating them produces the impression that flooding is unconditionally advantageous under depletion — true of the second argument, false of the first.

1.5.3 Placement on the spectrum

?? lays the release mechanisms out as a spectrum (??). The catalogue is built there and is not rebuilt here. What a per-delivery toll does at each station follows from the batch count without a new derivation: the surviving output is set by how many times ϑ is paid.

Under a per-delivery toll, selection on release phenotypes is therefore monotone toward fewer, larger deliveries (figure 1.12 and ??). That is a different statement from proposition 1.12, which compares two laws at one delivery each and can go either way. It is also what makes depletion a candidate selective pressure at all, in the sense in which nested within-host models pose the

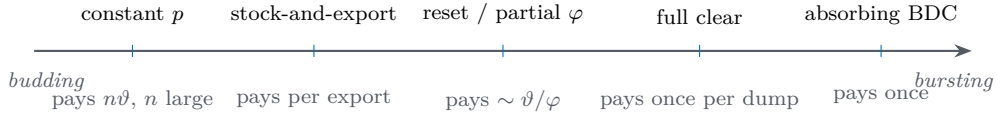


Figure 1.12: Per-delivery toll along the budding–bursting spectrum of $??$. Moving right, fewer deliveries are made and a smaller total is paid. The ordering is the batch count, not the distributional reversal.

question [31]: it supplies a gradient rather than merely a difference.

That gradient does not run unopposed. It runs against the one the chapter on population dynamics identifies, since concentrating release lengthens the generation time and so lowers the deterministic growth rate [32], and the phage lysis-timing literature poses the same trade-off in its own terms [33]; how a lytic strategy can be evolutionarily competitive at all under that tension is asked directly [11]. The contribution here is to name the third quantity in the trade-off and to give it a closed form.

Self-excitation ($??$) is the one station where the toll itself moves. Remark 1.5 assumes the killer field has recovered between deliveries; a self-exciting mechanism makes deliveries cluster in time, so successive dumps meet a field that has not recovered and pay less than ϑ each. Whether that is enough to overcome the batch-count penalty of making several deliveries instead of one is not settled here — it depends on the recovery timescale, which the model does not carry — and it is recorded as an open problem in section 1.6.2. It is worth naming because it is the only mechanism in the catalogue for which the two arguments of remark 1.13 pull in opposite directions.

This is a statement about which release phenotypes a depletable defence favours, not about which pathogens possess which phenotypes, and it holds at the level of one cell’s output. It inherits the restriction that $??$ and [10] carry: what is being described is the early phase [34].

1.5.4 *Y. pestis*

One reading takes the model of section 1.4 in the context of *Y. pestis* and neutrophils: the target cells are macrophages, the killer cells are neutrophils. *Y. pestis* does replicate within macrophages [35, 20], subverting the phagosome rather than being destroyed by it [36, 19], and it encounters those cells early in the pulmonary route [37]. The injected effectors of section 1.2.1 act on the same cells, driving or withholding apoptosis [18, 38, 39]. Both actors of the model are present. The formulae themselves do not depend on that reading.

Model	Reading
replicators X_t	intracellular pathogen in one macrophage
granules Q_t	remaining oxidative store
ς	suppression per replicator–granule pair
burst size r	load at rupture
budget ϑ	killer cells removed per delivery
$\mathcal{R}(\vartheta)$	expected surviving release

Table 1.4: A reading of already-named objects under the *Y. pestis* mapping. The formulae themselves do not depend on it.

Read that way, the argument is two-phase. While killer-cell density is roughly constant, the bursting advantage of section 1.5.3 applies: a macrophage releasing its whole load at once pays one toll where a budding cell would pay many. If a later rise in killer cells pushes the toll past $\vartheta^* = \log V_\infty / \log a$, no intracellular strategy remains supercritical (corollary 1.8), and the advantage lies with becoming extracellular. Why not replicate extracellularly from the beginning

is then a separate question: being within macrophages may buy transport to lymph nodes, and perhaps food itself.

What the mathematics supplies is a single number. At $(\lambda, \mu, \delta) = (1, 0.5, 1/3)$ one has $\vartheta^* = 1.71$, so a killer field able to remove two agents per delivery already forces the switch; at $(1, 0.2, 0.05)$ it is 44.13. If effector density at the site crosses ϑ^* during the infection, the intracellular phase should end at a level set by the intracellular rate constants and by nothing else. An analogous reading for *F. tularensis* [40, 24] would use the same threshold with different constants.

1.6 Discussion

1.6.1 Limitations

Of the three candidate advantages set out in [section 1.1.1](#), this chapter settles one. *Limited killing* is what [sections 1.3](#) and [1.4](#) model, and the results are the certainty threshold, the diagonal law and $\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$. The other two were named there and left, and this is where the reasons are given.

Subductive proliferation — a replication rate that declines during residence in a host cell — has its machinery already: it is a time-inhomogeneous birth process, a special case of ?? with $\lambda(t) = \beta e^{-\alpha t} + \gamma$, and the first-jump probability, the mean and the deterministic trajectory are all elementary. That material is not carried in this chapter, because it does not by itself distinguish the two release strategies. Indeed it is not obviously an argument for bursting at all: a rate that decays with residence time penalises precisely the strategy that waits before releasing, so the sign of the effect is unclear before the calculation is done, and the calculation is not done here. The apparatus is retained as a candidate worked example for ??, where the general result it specialises already lives.

Detection avoidance is named and not pursued because this chapter has no model of detection. It would need a hazard that responds to the rate of release rather than to its size, which is the opposite dependence from the one modelled here, and it is the mechanism most often invoked for HIV, where release is observed to be clustered and where the responding immune compartment has models of its own [41].

Four further restrictions are worth stating.

The replicator–suppressor process of [definition 1.1](#) lets the pathogen grow without bound once the store is spent. Real intracellular capacity is finite, and a full model would terminate the run at rupture rather than let X_t diverge. Nothing in [proposition 1.3](#) or [proposition 1.4](#) depends on the late-time behaviour — both are statements about which absorbing set is reached — but the survival maps of [section 1.3.4](#) should be read as probabilities of avoiding clearance, not as predictions of intracellular load.

The granule store is never replenished ([remark 1.2](#)). This is the assumption that makes depletion permanent, and it is what turns a larger cohort into a lasting advantage rather than a temporary one. With replenishment at any positive rate the certainty threshold would soften into a race between refill and consumption.

The removal budget is fixed, and [figures 1.9a](#) and [1.9b](#) show what is being idealised: a slowly saturating casualty count is replaced by a step. The replacement is what makes (1.18) available in closed form, and the honest claim is the one made in [section 1.4.1](#), that the qualitative dynamics are expected to be shared with the logarithmic case rather than that the two agree; which of a model’s predictions survive such a substitution is not a question that answers itself [42]. It also fixes ϑ as an integer ([remark 1.11](#)), which is right for a count of killed particles and wrong for an average over deliveries.

Finally, [proposition 1.12](#) compares single batches. It matches two laws at one delivery each,

exactly as ?? does, and [remark 1.13](#) sets out what that does and does not license. In particular it is not a statement about a whole infection, and the population-level consequences of a threshold clearance are not derived anywhere in this thesis.

1.6.2 Open problems

1. **The chained process at $\mu > 0$.** The joint laws of the rupture times and sizes (t_k, r_k) are known in closed form only for $\mu = 0$. ?? states why the decomposition fails once a lineage can vanish internally, and states the problem first; this chapter inherits it, because [section 1.2.1](#) is the model it concerns.
2. **Off the diagonal.** [Proposition 1.4](#) gives $\pi_{j,j}$ in closed form because level 0 of the first-step system closes on itself. No such product form is available for $\pi_{i,j}$ with $i < j$: the generating function in i couples level j to level $j - 1$ with $\pi_{1,j}$ entering as an unknown fixed by a degree condition. Off-diagonal values are computed and tabulated ([figures 1.6](#) and [1.7](#)) and left at that.
3. **The reversal at intermediate φ .** ?? asks where on the partial-release continuum of ?? the flooding advantage of ?? survives. [Proposition 1.12](#) answers the same question from the other side, at the two endpoints, and finds the criterion reversed. What happens at intermediate φ under a threshold is not known, and the two sub-questions are plainly the same problem viewed from opposite ends.
4. **Clustered deliveries and a recovering field.** If the killer field recovers on a timescale comparable with the interval between deliveries, then [remark 1.5](#) fails and the toll paid at successive deliveries is correlated. [Section 1.5.3](#) notes that self-excitation is the one mechanism in ?? for which this matters, and the question of whether reduced tolls can compensate for extra deliveries needs a model that carries the recovery timescale, which this one does not.

1.6.3 Connections forward

The selection gradient identified in [section 1.5.3](#) is a statement about variation between release phenotypes, and the chapter on variability in evolution is where variability itself becomes the object of study rather than a property of a fixed mechanism. The *Y. pestis* account of [section 1.5.4](#) is the piece this chapter contributes to the concluding chapter, whose subject is the evolved strategy of that organism: what is offered here is one testable number, the threshold budget ϑ^* , and the claim that the switch between the intracellular and extracellular phases should occur where the effector density crosses it.

1.A Numerical verification

Every quantitative claim in this chapter was checked against independent numerics before being stated. The checks were run in `numpy` with no other numerical library; the method in each case was to compute the quantity twice, once from the closed form and once from a construction that does not use it, and to compare.

Three of the rows in [table 1.5](#) deserve a word, because what they establish is used rather than merely tallied.

The reversal row is the strongest evidence in the chapter. The sign of $\mathcal{R}_{\text{burst}} - \mathcal{R}_{\text{bud}}$ was predicted from L alone, by [proposition 1.12](#), and compared against brute-force summation of both laws over ten rate regimes and five removal budgets. There were no exceptions among

Claim	Independent construction	Agreement
Burst law (1.13)	exact solve of the embedded jump chain on a 3,000-state truncation, cross-checked by Monte Carlo; 6 regimes	3.3×10^{-11} abs.
Normalisation $\sum_{i \geq 1} \frac{\delta}{\lambda} a^{-i} = 1 - b$	algebraic, then numeric	1×10^{-16}
$\mathbb{E}(r) = V_\infty$ (1.10)	direct summation to tail below 10^{-14}	1.3×10^{-16}
Upper tail (1.17)	partial sums of the burst law	1×10^{-16}
Partial-sum form (1.14)	brute-force summation; 7 regimes $\times$ 6 budgets	1.8×10^{-14} rel.
Intermediate form (1.19)	brute-force summation, integer ϑ ; 7 regimes	2.5×10^{-14} rel.
$\mathcal{R}(\vartheta) = V_\infty a^{-\vartheta}$ (1.18)	brute-force summation; 7 regimes $\times$ 8 budgets $\vartheta \leq 40$	3.1×10^{-15} rel.
Real- ϑ form (remark 1.11)	brute-force summation; 6 regimes $\times$ 9 budgets including 0, 0.5, 0.8, 1.5, 2.3, 4.7	2.1×10^{-15} rel.
Reversal (1.21)	brute-force summation of both laws; 10 regimes $\times$ 5 budgets, 50 rows	3.1×10^{-15} and 7.6×10^{-16} rel.; 0 sign mispredictions
First-step system (1.5)	400,000-path Monte Carlo of the embedded chain [43]; 12 initial states $\times$ 3 values of ς	every entry within ± 2 s.e.
Diagonal law (1.7)	full linear solve of (1.6) at $N = 25$; $\varsigma \in \{0.1, 0.2, 0.3, 1.0\}$	6.1×10^{-16}

Table 1.5: The verification record. The burst-law entry is the only one whose agreement is limited by anything other than floating-point arithmetic: its error is set by the truncation tail, not by the formula. Every other row agrees to within a few units in the last place, which is what one expects of two algebraically equivalent evaluations rather than of a formula that happens to fit.

the fifty rows. Swept in δ across the boundary at $\lambda = 1$, $\mu = 0.5$, the sign changes exactly at $\delta_* = 1/3$ to within 10^{-16} , which is the crossing drawn in figure 1.11.

The first-step row is the only stochastic check. The linear solve of (1.6) was compared against direct simulation of the embedded chain rather than against another algebraic route, because there is no second algebraic route off the diagonal. Monotonicity of $\pi_{i,j}$ in i and in j is a numerical check, not a claim of proposition 1.3: it was verified over the same grids at $N = 20$ and holds everywhere tested.

The intermediate-form row records a scope limit rather than an error. The expression (1.19) is exact at integer ϑ and is *not* the quantity $\mathbb{E}(\max(0, r - \vartheta))$ at non-integer ϑ , where its relative error reaches -14% at $(\lambda, \delta, \vartheta) = (0.8, 2, 0.8)$. That is the reason remark 1.11 declares ϑ integral and the reason the figures of section 1.4.3 are drawn at integer budgets.

The scripts that regenerate the figures of this chapter sit with the chapter source, one directory per figure, each self-contained and runnable on its own. The value $\mu = 0$ in the lifetime-yield and replication-ratio figures was recovered by fitting the original published curve; the verification computations above were run separately and are not reproduced here.

1.B Computational notes

1.B.1 The first-step solver

The system (1.6) is assembled directly from the jump probabilities of (1.5) and solved once. The unknowns are $\pi_{i,j}$ for $1 \leq i \leq j \leq N$; the two boundary conditions $\pi_{0,j} = 1$ and $\pi_{i,j} = 0$ for $i > j$ enter as the inhomogeneous term and as an omission from the index set respectively. The assembly is direct enough to describe in words, and the working implementation is a file rather than a printed page.

Index the transient states (i, j) with $1 \leq i \leq j \leq N$ in any fixed order. For each such state, place $\hat{\alpha}_j$ in the column of $\pi_{i+1,j}$ whenever $i + 1 \leq j$, and $\hat{\beta}_j$ in the column of $\pi_{i-1,j-1}$ whenever $i - 1 \geq 1$; when $i = 1$ the second coefficient has no column to go in, because $\pi_{0,j-1} = 1$ is known, and it becomes that row's entry of $\mathbf{c}$. Nothing else is non-zero. One call to a dense linear solver on $(\mathbf{I} - \mathbf{A})\mathbf{p}^{(0)} = \mathbf{c}$ then returns every $\pi_{i,j}$ at once. At $N = 20$ the system has 210 unknowns and solves in milliseconds, so the block-triangular structure of [figure 1.5](#) is a fact about the matrix rather than a computational necessity; it becomes one only for much larger stores.

Two features of that recipe are worth noting, and both come from [section 1.3.3](#). The coefficients depend on j alone, so each takes one argument where the state has two; and the omission of the states $i > j$ is not an approximation but [proposition 1.3](#), which is what keeps the system finite despite X_t being unbounded. Get either wrong and the solve still returns numbers, which is why the implementation checks its diagonal against the closed form of [proposition 1.4](#) on every run, and agrees to 1.9×10^{-16} .

The working implementation sits with the chapter's figure-generation scripts. It is self-contained, depends only on `numpy` and `matplotlib`, carries the assembly above, the solve, that check, and the plotting that produces [figures 1.6](#) and [1.7](#).

1.B.2 Quadtree sampling

Collision tests for the particle runs of [figures 1.1](#), [1.2](#) and [1.8](#) were quadtree-indexed, so that only agents local to a macrophage are tested. The index is rebuilt each frame; the saving is that collision testing stops being quadratic in the particle count, which is what the large-load runs of [figure 1.8](#) require.

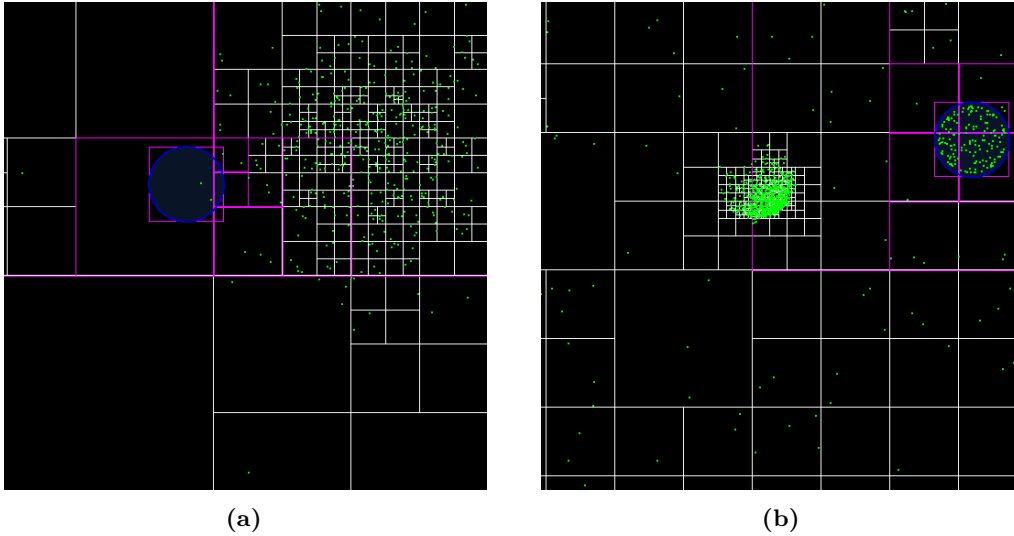


Figure 1.13: The quadtree index during a simulation run, with the cells local to the macrophage shaded pink; only the particles inside those cells are tested for collision. (a) A sparse field, where the index buys little. (b) A dense one, where it buys a great deal: the shaded region is a small fraction of the frame, so the number of collision tests grows far more slowly than the number of particles. This is why the large-load runs of [figure 1.8](#) are feasible at all.

[Figure 1.13](#) shows the index in a sparse field and a dense one.

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